

Assignment One: Python Basics

Edge computing device programming for AI projects

Student

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Deadline: 22/09/2026 11:59pm

GitHub link

GitHub link : <https://github.com/<YOUR-GITHUB-USERNAME>/assignment-one-python-basics>

Description

This is my submission for Assignment 1: Python Basics. All three tasks are implemented in one single Python file, assignment 1_WangZhihan.py, which is uploaded to the GitHub repository linked above. The program only uses the Python standard library, so it runs on the Jetson Nano with no extra installation. When the file is started it shows a small menu, and each task can be opened on its own or all three can be run one after another.

Task A - Simple Calculator

The calculator asks for the first number, the second number and then the operator (+, -, *, /), and prints the result in a clear form such as "Result: 3 + 5 = 8". It also accepts a whole expression typed on one line, for example 3 + 5. Input errors are handled instead of crashing the program:

- dividing by zero prints "Cannot divide by zero"
- an operator other than + - * / prints "Invalid operation"
- letters typed where a number is expected are asked for again
- a line that is not a proper expression is reported and asked again

Task B - Question Answering Bot

The bot receives the question with input() and chooses the reply with an if / elif / else chain. It supports the five required keywords:

```
hello    -> Bot: Hello! Nice to meet you.
python   -> Bot: Python is a language, and it is very friendly to beginners.
jetson   -> Bot: Jetson Nano is an AI computer made by NVIDIA.
ai       -> Bot: AI means Artificial Intelligence.
name     -> Bot: My name is Python Bot.
```

The matching ignores capital letters and punctuation, so Hello!, PYTHON? and "what is AI" are all understood. Any other question is answered with a friendly "Sorry, I don't understand.", and typing bye ends the conversation.

Task C - Turtle Drawing

The drawing task uses turtle with for loops, changing colours and changing lengths. It can draw:

- a square, a triangle and a five-pointed star
- a colourful rotating pattern that uses six colours and shrinks each round
- a spider web with twelve spokes in eight different colours

The five pictures can be drawn one by one or all together. When the drawing is finished, clicking the window returns to the menu.

Testing

I tested the program before submitting. The calculator was checked with all four operators, with decimal numbers, with a division by zero and with wrong input; the bot was checked with the five keywords in upper and lower case and with unknown questions; and the turtle drawing was checked by saving the drawing and confirming that all five pictures are produced. The program did not stop with an error in any of these tests.

File submitted: assignment 1_WangZhihan.py Report: Assignment1_WangZhihan.pdf